

Quantum Computing Research Portfolio

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Quantum Information Theory, Complexity, and AI

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github.com/asmeeta-quantum/ETH_Qiskit_Workspace

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Abstract

This portfolio synthesizes my independent research across three interconnected domains: (1) quantum foundations and information theory, (2) quantum complexity and verification, and (3) quantum computing and artificial intelligence. My work combines analytical modeling of open quantum systems with numerical simulations using Qiskit and QuTiP, alongside exploration of AI-driven methods for quantum algorithm discovery and error mitigation. Background at ISRO provides practical insight into physical noise sources in cryogenic and microwave systems relevant for quantum devices.

Research Goal: Develop rigorous frameworks for understanding the fundamental limits of noisy quantum computation, with applications in verification, error mitigation, and quantum-enhanced AI.

Research Domains

Domain 1: Quantum Foundations and Information Theory

- Quantum information theory and open quantum systems
- Noise characterization and quantum channel divergences
- Verification complexity under realistic noise
- T_1/T_2 relaxation and decoherence analysis

Domain 2: Quantum Complexity and Verification

- Complexity-theoretic analysis of quantum advantage
- Classical simulability boundaries under noise
- Verification protocols and sample complexity
- Resource theories for noisy quantum computation

Domain 3: Quantum Computing and Artificial Intelligence

- AI-driven quantum error mitigation and optimization
- Hybrid quantum-classical machine learning
- Quantum-enhanced AI for combinatorial optimization
- Neural networks for noise characterization

Key Research Contributions

Verification Complexity Under Noise

Objective: Understand how T_1/T_2 relaxation and dephasing affect verification sample complexity.

Methods:

- Analytical scaling bounds for verification under noise channels
- Quantum channel divergences and entropy measures
- Identification of noise thresholds for classical simulability

Key Result: Derived analytical bounds identifying regimes where quantum advantage becomes classically simulable.

Error Mitigation Theory and Simulation

Objective: Develop and validate error mitigation frameworks for noisy quantum circuits.

Methods:

- Zero-Noise Extrapolation (ZNE) implementation
- Systematic noise amplification analysis
- Benchmarking on simulated noisy circuits

Key Result: Demonstrated $\approx 20\%$ fidelity improvement on noisy quantum circuits.

AI-Informed Quantum Control Optimization

Objective: Suppress non-computational excitations in weakly anharmonic transmon qubits using AI-driven optimization.

Methods:

- DRAG pulse optimization framework
 - Lindblad master equation simulations
 - Leakage-aware control theory
- Key Result:** Achieved > 90% suppression of $|1\rangle \rightarrow |2\rangle$ leakage.

Hardware-Informed Noise Modeling

Objective: Translate microwave component stability data into quantum noise parameters.

Methods:

- Phase noise characterization from VNA measurements
- Cryogenic system stability analysis (4K)
- Composite noise channel modeling

Key Insight: Hardware characterization data provides practical constraints for quantum noise models, bridging theory and experiment.

Research Portfolio Structure

ETH_Qiskit_Workspace/

- 1. 01_bell_states_entanglement.ipynb
- 2. 02_statevector_simulation.ipynb
- 3. 03_depolarizing_noise_analysis.ipynb
- 4. 04_superconducting_noise_channels.ipynb (T₁/T₂ modeling)
- 5. 05_composite_noise_simulation.ipynb
- 6. 06_grover_verification.ipynb
- 7. 07_deutsch_jozsa_advantage.ipynb
- 8. 08_verification_sample_complexity.ipynb (Key: analytical bounds)
- 9. 09_lindblad_qubit_dynamics.ipynb
- 10. 10_drag_pulse_optimization.ipynb (Key: > 90% suppression)
- 11. 11_noise_bench_qc/ (Benchmarking suite)
- 12. 12_lindblad_sim/ (Simulation toolkit)
- 13. 13_ai_error_mitigation/ (AI-driven ZNE)
- 14. 14_hybrid_qml/ (Hybrid quantum ML)
- 15. 15_qaoa/ (QAOA optimization)

Technical Implementation

Software Tools Developed

Tool	Description	Key Result
Noise-Bench QC	Benchmarking suite for verification under realistic noise	Systematic noise characterization framework
Lindblad-Sim	Non-Markovian and leakage-aware simulation toolkit	Open system dynamics simulation
ZNE Toolkit	Zero-Noise Extrapolation framework	≈ 20% fidelity improvement
DRAG Pulse Optimizer	Adaptive pulse design for leakage suppression	> 90% leakage suppression
AI-Error Mitigation	Neural network-driven error mitigation optimization	Automated ZNE parameter tuning
Hybrid-QML	Hybrid quantum-classical ML framework	Quantum-enhanced feature extraction

Mathematical Methods

- **Smooth Entropies:** Quantifying information leakage and verification complexity

- **Quantum Channel Divergences:** Characterizing recoverability and noise thresholds
- **Lindblad Dynamics:** Modeling composite noise evolution
- **Resource Theories:** Formalizing noisy quantum advantage
- **Tensor Networks:** Decoding and error correction

Key Achievements

1. Analytical bounds linking physical noise (T_1/T_2) to verification scalability
2. Open-source tools for realistic quantum noise simulation
3. Hardware-to-quantum modeling pipeline from ISRO experience
4. Identification of noise-triggered simulability transitions
5. $> 90\%$ leakage suppression via DRAG pulse optimization
6. $\approx 20\%$ fidelity improvement via zero-noise extrapolation
7. AI-driven frameworks for quantum error mitigation
8. Hybrid quantum-classical ML models for optimization

Repository and Code Availability

Full Repository: https://github.com/asmeeta-quantum/ETH_Qiskit_Workspace

All notebooks, simulation toolkits, benchmarking suites, and AI/ML frameworks are available for review.